

Learning to Build Momentum: Experience Replay, On-Policy Learning, and Reward Shaping in MountainCar

Abstract

MountainCar tests whether an agent can learn that reaching a goal to its right requires first moving away to build momentum. We compared a random policy, tile-coded SARSA(λ), DQN with and without replay, replay DQN with potential-based energy-deficit shaping, and a replay DQN with premature exploration decay. Each learned condition used five fixed training seeds and 60,000 interactions; validation selected checkpoints before greedy evaluation on 20 held-out seeds per policy. SARSA(λ) was strongest: it completed every held-out episode and reached the flag within 100 steps in 0.380 [0.330, 0.420], which was not reliable satisfaction of the target. Replay did not improve the locked DQN outcome. Shaped replay recorded occasional discoveries and 0.020 [0.000, 0.060] final assignment success, but no robust improvement. Rapid decay did not produce the registered aggregate valley-trapping signature. The principal limitations are five training seeds, a modest budget, and different function approximators for SARSA and DQN.

1. Problem

MountainCar-v0 has a two-dimensional continuous state, position (x) and velocity (v), and three discrete actions: accelerate left, coast, or accelerate right. An episode normally lasts at most 200 steps. We distinguish environment completion—reaching the flag within that limit—from the stricter assignment success criterion of reaching it within 100 steps.

The car's engine is too weak to drive directly up the right hill from the valley. Always accelerating right therefore settles into an unproductive motion. A successful controller alternates acceleration with the direction of travel, climbs one slope, reverses, and converts height into speed. This oscillation increases the amplitude of successive passes until the car can cross the goal at ($x=0.5$). Moving left can thus be useful even though it temporarily increases distance to the flag.

2. Methods

The random policy was a no-learning sanity check. SARSA(λ) represented the continuous state with 16 overlapping 8×8 tile codings and used replacing eligibility traces. For active tile features, its on-policy update used

$$\delta t = r_t + 1 + \gamma Q(st+1, at+1) - Q(st, at),$$

with epsilon-greedy behavior, $\gamma = 0.99$, $\lambda = 0.9$, and the configured learning rate divided across tilings. True goal termination removed the bootstrap term; artificial time-limit truncation did not.

All DQN conditions normalized observations from the environment bounds and used the same two-layer 64-unit ReLU network, Adam optimizer, Huber loss, gradient clipping, and periodically copied target network. Their target was

$$yt = r_t + \gamma(1 - \mathbb{1}[\text{terminated}_t]) \max_a Q_{\text{target}}(st+1, a).$$

Online DQN updated on the current transition with batch size one. Replay DQN sampled uniform batches of 64 after a 1,000-transition warm-up from a 50,000-item buffer. Removing replay therefore also removed minibatch decorrelation and changed batch size; this is not a memory-only intervention.

The shaped replay condition changed only the training reward. Its interpretable, dimensionless energy proxy—not exact simulator energy—was

$$E(x, v) = 0.45 \sin(3x) + 0.55 + \eta \text{clip}(v/v_{\text{max}}, -1, 1)^2, \Phi(s) = -\text{clip}(E_{\text{target}} - E(s)E_{\text{scale}}, 0, 1),$$

and the learner received $r' = r + \beta[\gamma\Phi(s') - \Phi(s)]$, with $\beta = 0.5$, $\eta = 0.5$, and $E_{\text{scale}} = 0.9$. Squared velocity is direction-neutral, so leftward momentum is not penalized merely for increasing goal distance. Potential was set to zero at true termination and retained at time-limit truncation. Final evaluation always used the original reward.

The deliberate failure condition was valid replay DQN with epsilon decaying from 1.0 to 0.05 over 200 interactions rather than 20,000, placing it near its minimum before replay warm-up. No code was broken and the condition was not redefined after seeing its outcome.

3. Experimental protocol

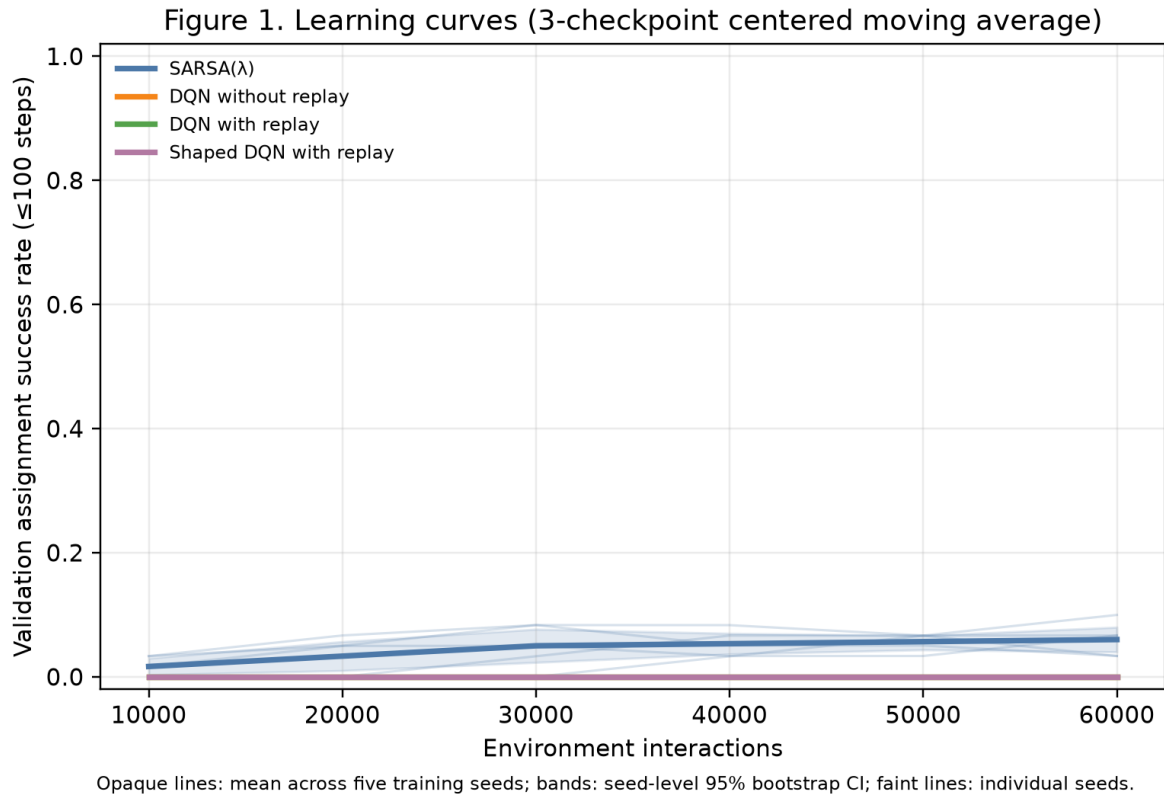
Training seeds were fixed at 0–4. Greedy validation used seeds 1000–1019 at 10,000-interaction intervals. After bounded validation and configuration locking, the selected checkpoint for each run was evaluated on disjoint held-out seeds 2000–2019. Every learned run received exactly 60,000 interactions (300,000 per condition and 1,500,000 across the five learned conditions); SARSA also recorded its episode count. The random baseline did not learn.

Checkpoint selection was lexicographic: highest validation assignment-success rate, highest validation completion rate, lowest median completed length, lowest episode-length standard deviation, then earliest checkpoint on an exact tie. Neither configuration nor checkpoint selection used final-test episodes.

Evaluation stopped on `terminated` or `truncated`, counted actual `env.step` calls, and was greedy. A 101–200-step completion was never counted as assignment success. Reported 95% intervals are percentile bootstrap intervals from 10,000 deterministic resamples of the five training-seed summaries. The 20 evaluation episodes within a policy were not treated as independent training replicates.

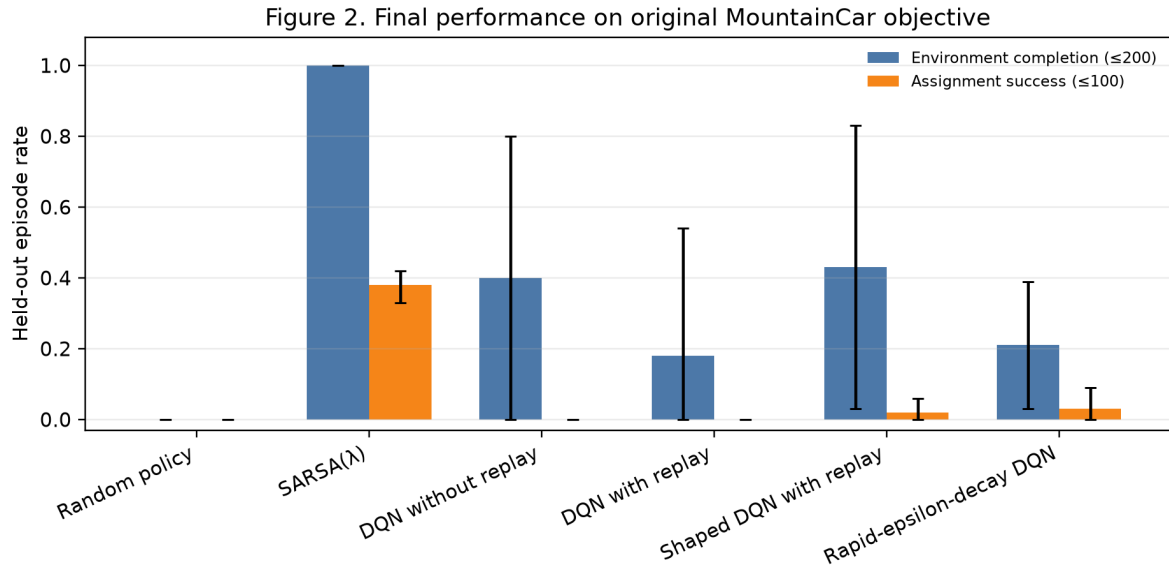
4. Results

The complete seed-level summary is [Table 1](#). Random never completed and returned -200.000 [-200.000, -200.000]. SARSA(λ) completed 1.000 [1.000, 1.000] of held-out episodes and achieved assignment success 0.380 [0.330, 0.420], with median completed length 104 steps and mean original return -100.940 [-102.840, -99.310]. Its across-seed assignment-success standard deviation was 0.057. This was the strongest measured result, but a 38% rate does not satisfy a reliability claim.



Learning curves

Figure 1. Greedy validation assignment success against interactions. Opaque lines are five-seed means, bands are seed-level bootstrap intervals, faint lines are individual seeds, and the three-checkpoint centered moving average is for display only. Episode-boundary overshoots were linearly aligned to the fixed 10,000-interaction validation grid before smoothing. SARSA shows a small late increase; no DQN shows sustained within-100 success at this budget.



Error bars are 95% bootstrap intervals across five training seeds.

Final performance

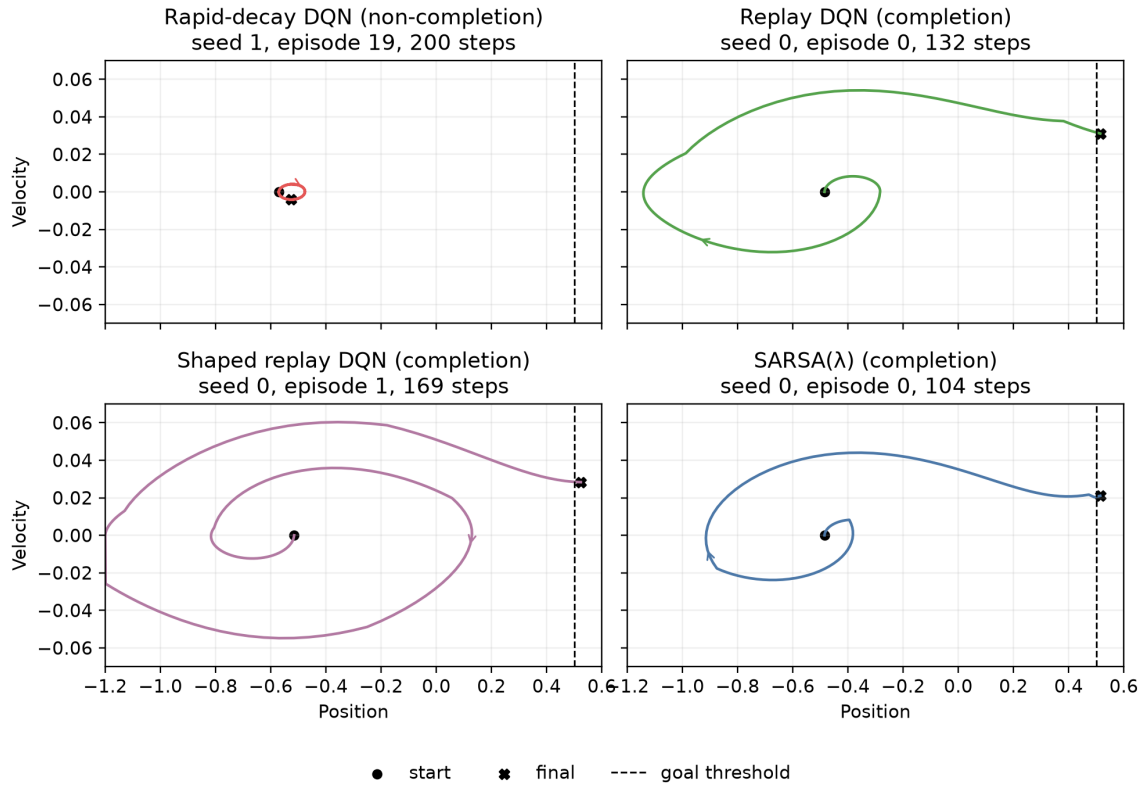
Figure 2. Original-objective held-out completion and assignment success, with uncertainty across training seeds.

Online DQN completed 0.400 [0.000, 0.800], compared with 0.180 [0.000, 0.540] for replay DQN; both had zero held-out assignment success. Their median completed lengths were 151.5 and 132.0 steps. Replay’s completion dispersion was lower (seed SD 0.402 versus 0.548), but its mean completion and original return were worse: -187.840 versus -177.610 . Standard replay observed no training completion in any seed, while online DQN’s median first completion was 31,780 interactions (observed range 26,516–46,355). Thus the predicted replay advantage in sample efficiency or stability was not established.

Shaped replay completed 0.430 [0.030, 0.830] and achieved assignment success 0.020 [0.000, 0.060], with median completed length 169 and mean original return -183.470 [-199.320 , -163.210]. Its median observed first training completion was 19,196 interactions (observed range 14,784–23,146), but two of five seeds were censored. Only one seed recorded a within-100 training success, at 59,161 interactions. Shaping was associated with occasional discovery and did not show material final-objective harm relative to standard replay, but the sparse events and broad intervals do not demonstrate reliable acceleration or improvement.

5. Behavioral and failure analysis

Figure 3. Position-velocity trajectories (identical axes)

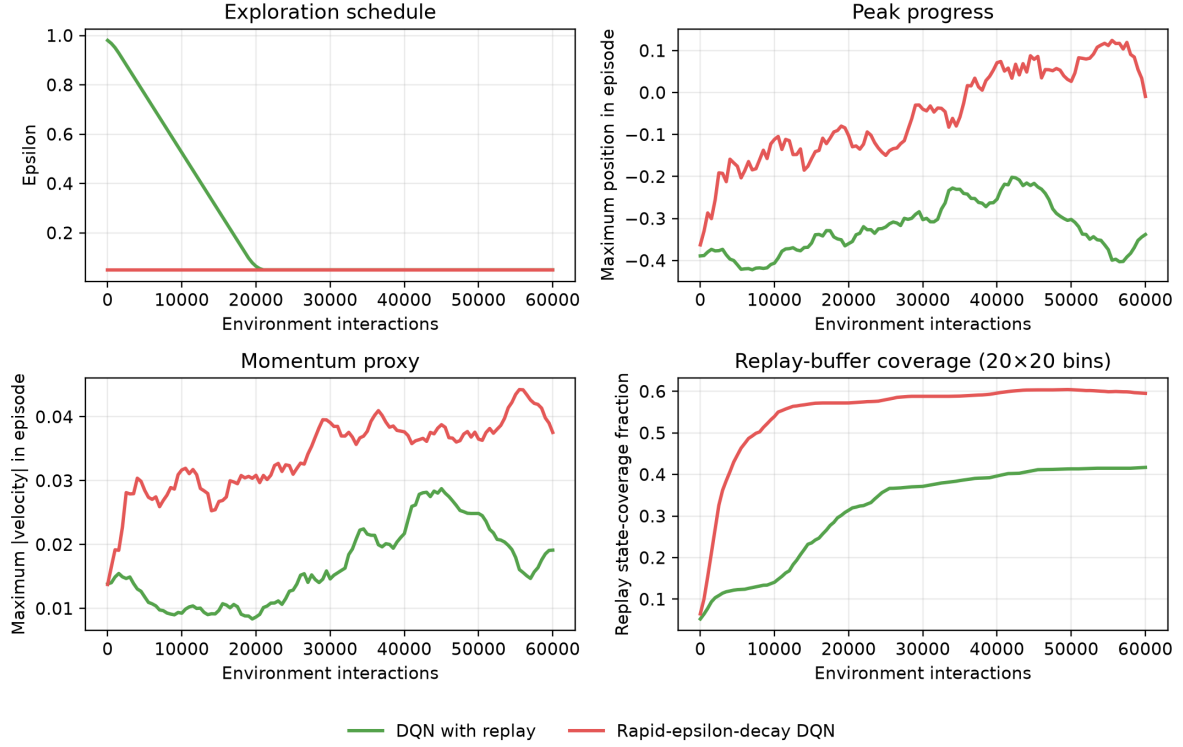


Position-velocity phase space

Figure 3. Deterministically selected held-out trajectories on identical axes. The successful examples were closest to each condition’s pooled median completed length; “completion” here does not imply the 100-step target. The rapid-decay non-completion was closest to the failed-episode median maximum position.

Replay DQN, shaped replay DQN, and SARSA trace expanding loops in position-velocity space. Each reversal permits a wider subsequent swing and ultimately a crossing of the goal threshold. The shown rapid-decay episode instead makes a tiny loop near its start and ends after 200 steps, illustrating one way a nearly greedy policy can fail to accumulate momentum.

Figure 4. Replay DQN versus rapid-epsilon-decay diagnostic traces



Failure diagnostics

Figure 4. Five-seed training diagnostics for standard and rapid-decay replay DQN. Curves are interpolated to a 500-interaction grid and use a five-point centered display smoother.

The aggregate evidence contradicts a simple valley-trapping account. Rapid decay reaches minimum epsilon almost immediately, yet it shows broader replay state coverage, larger peak position, and greater maximum absolute velocity than standard replay through much of training. It also obtained 0.030 [0.000, 0.090] held-out assignment success and 0.210 [0.030, 0.390] completion. Consequently, the preregistered hypothesis that premature exploitation systematically prevents momentum discovery is not supported. The selected failure trajectory is real but not representative evidence for a condition-wide mechanism. Possible alternative explanations include unstable value optimization, seed-dependent action preferences, weak checkpoint transfer to held-out starts, or simply insufficient interaction budget; these diagnostics do not distinguish among them.

6. Limitations

Five training seeds permit only coarse uncertainty estimates, especially for rare successes. MountainCar is a small deterministic benchmark, so conclusions may not transfer to noisy, high-dimensional tasks. Tuning was intentionally bounded and the 60,000-interaction budget was practical rather than exhaustive. SARSA and DQN differ in tile-coded versus neural approximation, eligibility traces, optimization, and policy regime; their difference is not a controlled on-policy/off-policy effect. The replay ablation is more controlled, but also changes batch size and temporal correlation. Finally, the shaping quantity is an approximate energy proxy rather than exact mechanical energy, and potential-based theory does not guarantee equal finite-training behavior with function approximation and checkpoint selection.

7. Conclusion

No condition reliably met the ≤ 100 -step assignment target. SARSA(λ) was the clearest learner at this budget, completing every held-out episode but meeting the strict target only 38% of the time. Replay did not improve this DQN implementation over online updates on the measured endpoints. Energy-deficit shaping was associated with occasional earlier completion and a small nonzero final success rate, but not robust benefit. The planned rapid-decay failure did not exhibit its predicted aggregate mechanism. The narrow empirical lesson is that successful policies used oscillatory momentum-building trajectories, while neither replay, shaping, nor an intuitive exploration story earned a general causal claim from these five-seed results.

8. Reproducibility

Run from the repository root with Python 3.12:

```
# Install the locked environment.
uv sync --frozen --python 3.12 --extra dev

# Run tests and all short smoke conditions.
bash scripts/smoke_test.sh

# Reproduce all locked training runs (expensive).
bash scripts/run_full_experiments.sh

# Evaluate existing locked checkpoints on the frozen held-out seeds.
uv run --python 3.12 mountaincar-run-suite \
  --evaluate-only --output results/processed/final_manifest.json

# Regenerate the validated table and four figures without training/evaluation.
uv run --python 3.12 python -m mountaincar_rl.analysis.make_all_figures
```

Raw metrics, configuration snapshots, and evaluation trajectories are under `results/raw/full/`; locked configuration hashes are in `configs/full/locked_manifest.yaml`.